Intented: phase 2

What is done after phase 1 (1–6 November) and the plan for phase 2 (7–14 November). For Furqaan and William. 6 November 2026, from `main` at `714330a`.

> **Where we are.** The whole flow runs on NEAR testnet. An agent proposes a payment, a model in a TEE on NEAR AI Cloud judges it against the owner's mandate and signs the verdict, and the wallet contract pays only on a valid signed `ALLOW` within its caps. The honest payment and the attack, both NEAR-native and cross-chain to Base, have transaction hashes. Anyone can make a wallet through the factory. The API holds no NEAR key. All 369 tests pass.
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> **What phase 2 is.** Integration in the UI, the loose ends below, an optional Claude Code and Cursor integration (MCP server and skill, still to be agreed), mainnet on 11 November, then the video, the freeze and the submission.

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1. What is done

Piece	Owner	State	Tests
Kill tests	Both	**KT1, KT2, KT3 passed.** A NEAR AI signature is verified on chain with `env::ecrecover`, and Qwen3.8-27B was chosen over GLM. KT4: the dry 1Click quote works; **the real $2 payment is not done** and waits on mainnet.	
Wallet contract `contract/wallet`	Furqaan	`propose` and `execute` with every check: an allowlisted signer over the exact request and response, `ALLOW`, the exact action and args, mandate and prompt hashes, the converted amount, nonce, expiry, per-token and NEAR caps per UTC day, freeze, and the cross-chain recipient list and 1Click's signed quote. At most 32 proposals pending, expiry at most 1 h ahead, `prune`, `withdraw`, agent keys, `new_with`, in-place migrations. Live at `intented-wallet.testnet`.	57
Wallet factory `contract/factory`	Furqaan	Live at `intented.testnet`. `create` makes `<name>.intented.testnet` in one batch, with the caller as owner and no full-access key. It refunds the deposit on failure. `get_wallets` per owner. Code hash `Gy6sretT…`; `set_code` only by the factory account.	9
Firewall core `firewall-core`	Furqaan	`judge`, the canonical encoder, the prompt with untrusted data quoted, a fail-closed verdict parser, the converted token amounts, the `cross_chain` field, the signature fetched straight after the call, and the seam (`proposeOnChain`, `executeOnChain`), now used by the agent.	141
Attestation	Furqaan	`pnpm attest`: the Intel TDX quote via dcap-qvl, report_data bound to the signer and a fresh nonce, and NVIDIA's GPU JWT. The signer `0x835ab53d…aa26` is verified, published in `docs/attestations/` and allowlisted. Done off chain; the contract does not verify the quote.	(in core)
API `api`	William	`/api/mandate`, `/api/mandates/:hash`, `/api/quote`, `/api/verdict`, `/api/executed`, `/api/proposals`, `/api/redteam/latest`. It judges, quotes 1Click and reads the chain. **It holds no NEAR key**: `/api/propose` and `AGENT_PRIVATE_KEY` were removed on 5 November. Rate limits on verdicts (30 per wallet per hour) and quotes (30 per wallet, 120 in all, per hour).	90

Agent and paywall agent	William	`intented init` (the agent makes its own key; the Create link) and `intented pay`. The four demos (honest, poisoned, base, base-poisoned) propose and execute with the agent's own key. A mock x402 paywall and the testnet 1Click stand-in.	56
Web app web	William	Landing, Activity (a receipt per payment: reason, signer, signature, attestation link, transaction, including the refused `execute`), Rules (the mandate checked against its on-chain hash), Create, My wallets, and a wallet page with the owner's controls and a payments chart.	none
Red team redteam	William	83 cases × 3 runs through the real `judge()` on 5 November: **44/44 attacks blocked** (12 aimed at the judge, 8 cross-chain), **0/39 honest payments stopped**, 0 errors in 249 runs, all signed by the allowlisted address. **6.0 s median, over the 5 s bar.**	16
Cross-chain to Base	Both	Works on testnet with the 1Click stand-in. The judge sees the Base recipient, token and amount; the wallet checks the recipient list and 1Click's ed25519 quote signature. **Delivery is simulated on testnet**; real 1Click runs only on mainnet.	

Proof on testnet

What	Wallet	Transaction
Honest payment, 0.001 wNEAR to the vendor	intented-wallet	`C8P7yHvwT6e6C5LZLgXQ4uXDJwyoCDGX6Y7Vq5uvbtHH`
Attack refused: 0.01 wNEAR to the attacker, under the cap ("verdict is not ALLOW")	intented-wallet	`FvsDHLYMAXuAuTSUyDzhUzG3Z9kPnzv4FD9MxDj88jxm`
Base payment, 0.4 USDC to the vendor's Base address (simulated delivery)	intented-wallet	`6ADu75vEGwK8tKoPbw7Nfmh7BwTjJfX151WRYvDY5Vhg`
Base attack refused by the recipient list	intented-wallet	`DbLMFYWsUf4r9DXAfnHfKf1NVyqacM9EVCyeb8nq3W5U`
Base attack refused on the verdict alone, list empty	intented-wallet	`GYQKNHhZbhxrqNzi8rYs7wJaoD3Hif4LSKZVDFXYfn8P`
Factory wallet: honest paid / poisoned refused	demo.intented	`HNRq1HiXRVgPyevDKBZgRsEZQM9xDjRKRmY4aF7fPqw9` `9WAKGzuMq6EGoh3rdUJj9Z71SdYr91tJRiJ9C6YVa7A6`
Bounded-proposals code: honest paid / poisoned refused	demo2.intented	`2eebywtFD8vU67C9NS3DSdi6dQRpUYVHSS6ZB1FD4ayp` `6bB1fqJFBSdcPECbNw2BQHmKmDFHusZE1FCyV1WX8H4C`
`intented pay`, the agent's own key: ALLOW paid / injected DENY refused	retest-1104.intented	`4rddy8CUvMfzJjxVALZdMPNmZ7W8ofKB5KERee66X2og` `62h7SnsLzY9vmXZj2Nhu3roHhRDvSEuYi8WVqM1fcVPC`

What is verified and what is trusted

What the README says now, which phase 2 must not overstate:

- **Attestation is checked off chain**, then the signer is allowlisted. It does not compare the app image with a known-good one, since none is published.
- **The model can be wrong.** The red team uses our own 83 cases, not an adaptive attacker. A compromised agent can re-propose and get up to 30 verdicts an hour per wallet. The caps bound what one wrong `ALLOW`

costs.

- **1Click is trusted to deliver.** The wallet verifies the signed quote, not the delivery on Base.
- **The team holds `intented-wallet.testnet`'s full-access key.** Factory wallets have none, and the mainnet wallet's key is deleted after setup.
- **The agent chooses what it passes as context.** The mandate check and the caps hold regardless.

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2. Loose ends before phase 2

Item	Owner	Next step
Factory and bounded-proposals rows are not in the interface spec, though the code is live	William's yes	On his yes Furqaan copies `docs/note-to-william-2026-11-04-spec-rows.md` in, with an acceptance record
Was `AGENT_PRIVATE_KEY` ever set on a hosted environment?	William answers	If yes, Furqaan replaces the key on `intented-agent.testnet`: one add-key, one delete-key
Stale docs: README factory test row (6, now 9); status board (wallet 31 → 57, core 126 → 141, no factory row); firewall-core README calls the seam "the seam William's API imports"	Furqaan	A small fix, one commit
Two records to confirm: "quote skips the recipient list" says "agreed by Furqaan"; a spec revision row (refused `execute` on `/api/executed`) lists William only	Furqaan	Confirm, or write a superseding record
OutLayer (open Discord question 3); first on the cut list	Furqaan	Recommend: cut now, record it
Other open Discord questions: 1 (signature retention, rotation), 4, 6, 8; William's 5 and 9	Both	Only 1 affects the build (re-check the signer before mainnet); the rest can close as not needed
Latency: 6.0 s median against the 5 s bar; the slow tail is the model call	Both	Report it honestly, as the README does; no prompt change this late without a red-team rerun

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3. Phase 2 plan

Day	Work	Gate
Sat 7 Nov Integration day	The full flow in the UI on testnet: a new wallet made from the Create page with `intented init`'s key, rules set, an honest payment and the attack, receipts in Activity. Fix the loose ends above. Settle the MCP decision (section 4).	The flow works from a fresh browser
Sun 8 Nov	If agreed: build the MCP server and the skill, one tool at a time. Furqaan writes the mainnet runbook.	
Mon 9 Nov	MCP check by hand in Claude Code and Cursor; record a coding agent tricked into paying and the wallet refusing. Read-only mainnet checks: re-run `pnpm attest` (signer unchanged?), verify a live mainnet 1Click quote against `ed25519:reYaWh…`, compute the mainnet prompt hash.	Checks pass, or a fix is planned
Tue 10 Nov	**Feature-complete on testnet.** Red-team rerun only if the prompt or model changed.	Nothing new after today but fixes

Wed 11 Nov	**Mainnet**, following the runbook: `deploy–wallet.sh`, configure, register the attested signer, set 1Click's real key, fund within the $20–50 budget, one allowed and one refused payment, the real $2 1Click payment (KT4), then delete the full-access key.	**A yes from both owners before any step**
Thu 12 Nov	Video and X post (William). README final pass with mainnet hashes; every claim checked against the code (both).	
Fri 13 Nov	**Feature freeze.** Fresh-clone test: clone, install, run every test and one demo from the README alone.	A stranger could follow the README
Sat 14 Nov	**Submit once**, by 20:00 IST (hard deadline 23:59 UTC). Furqaan fills in the form.	Irreversible

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4. Proposed: Intented for Claude Code and Cursor

PROPOSED Not yet agreed. Claude Code and Codex are among the judges, so this would let a judge's own coding agent pay through Intented, and it shows the problem where it happens: a coding agent reads a poisoned README or issue and tries to pay.

What it is

- **An MCP server** (stdio) for Claude Code, Cursor and Codex:
  - `intented_init(name)`: the agent's key and the Create link
  - `intented_pay(name, to, amount, token?, context)`: propose, verdict, execute
  - `intented_status(name)`: caps, today's spend, last proposals
- **A Claude Code skill** that teaches the coding agent to pass what it read as `context` and never to reword a payment after a `DENY`. The same text works as a Cursor rule.
- A thin wrapper over `agent/`'s `initAgent`, `loadAgentKey` and `pay()`. Contract, core and API unchanged.

To settle first

- **Lane.** It wraps William's code. Options: William builds it in `agent/`; Furqaan builds it in a new `mcp/` folder with William's agreement (recommended); or Furqaan builds it in `agent/` with William's OK.
- **Dependency.** `@modelcontextprotocol/sdk`, replacing nothing, pinned to a release at least two weeks old.
- **Keys.** The key stays in `~/.intented/`; no tool result ever returns the private key.
- **Honesty.** The README says the coding agent chooses its context; the mandate and caps still hold.
- **Setup for judges.** Run from a clone (nothing on npm) with their own testnet wallet.
- **Cost.** About a day. It is cut first if 10 November is tight.

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5. Risks

Risk	What we do
The deadline	The cut order: the MCP (new, proposed) → OutLayer → the mandate editor UI → the daily cap → the cross-chain leg (keep the NEAR-native payment). **The attack demo is never cut.**
Real money on mainnet	Only after both owners say yes, only $20–50 in all, every step from the runbook. The full-access key is deleted once setup is checked.

The real 1Click path is untested	The quote signature is checked against a live mainnet quote on 9 November; the $2 payment on 11 November. If it fails, the README says so and the NEAR-native payment carries the demo.
The signing key rotates	Re-attest before mainnet. A new signer is attested, published and allowlisted the same way.
Claims outrun code	Each README line is checked against the code and a hash before freeze. AI judges read it.
Secrets in a public repo	Keys stay in the password manager and `~/.intented/`; `.env` stays gitignored. Check `git log -p` for keys before submitting.

Sources: the build plan (`docs/Intented hackathon plan.pdf`), `docs/decisions/` (39 records), `README.md`, `docs/README.md`, and the test runs on 6 November: wallet 57, factory 9, firewall-core 141, api 90, agent 56, redteam 16, all passing.